

Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

Florian Rzepka^{a,*}, Qinnan Chen^a, Julia Kowal^a

^a*Technical University of Berlin, Chair of Electrical Energy Storage Technology, Einsteinufer 11, Berlin, 10587, Germany*

Abstract

This paper presents a measurement-only performance benchmark for embedded battery state estimation under realistic Battery Management System (BMS) constraints. The study isolates estimator performance rather than hardware-resource or implementation-effort evaluation and compares four embedded-ready estimators on the same Lithium Iron Phosphate (LFP) full-cell trajectory under a shared embedded constraint: a direct-measurement model (DM), a hybrid direct-measurement model with learned State-of-Health support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with GRU-based SOC estimation and LSTM-based online SOH context (DD). The benchmark perturbs only measurable inputs and timing information, including ADC quantization, current bias, additive current, voltage, and temperature noise, missing samples, irregular sampling, burst dropout, voltage spikes, and initial-state mismatch, so that all disturbances remain physically interpretable and reproducible. The comparison reveals no universal winner across deployment-relevant dimensions. Under clean conditions, HDM provides the lowest baseline error with an MAE of 0.0024, followed by HECM with 0.0090,

*Corresponding author.

Email addresses: florian.rzepka@tu-berlin.de (Florian Rzepka), qinnan.chen@campus.tu-berlin.de (Qinnan Chen), julia.kowal@tu-berlin.de (Julia Kowal)

DD with 0.0100, and DM with 0.0311. Under disturbed measurements, HECM is the strongest overall robustness choice, especially in the most discriminative current-bias and voltage-noise scenarios. DD is the most stable estimator under missing samples and recovers fastest after initialization mismatch, while DM shows the expected drift sensitivity and HDM trades nominal accuracy against reduced disturbance tolerance. The benchmark therefore exposes complementary failure modes of direct-measurement, hybrid physics-based, and data-driven estimators and supports decision-oriented model selection for embedded BMS deployment.

Keywords: Battery management system, State of charge, State of health, Performance benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

1 Nomenclature

2 *Acronyms*

3	BMS	Battery management system
4	SOC	State of charge
5	SOH	State of health
6	DM	Direct-measurement model
7	HDM	Hybrid direct-measurement model
8	HECM	Hybrid equivalent-circuit model
9	DD	Data-driven neural model
10	ECM	Equivalent circuit model
11	EKF	Extended Kalman filter
12	MAE	Mean absolute error
13	RMSE	Root-mean-square error
14	ADC	Analog-to-digital converter
15	CV	Constant-voltage phase

16	OCV	Open-circuit voltage
17	GRU	Gated recurrent unit
18	LSTM	Long short-term memory
19	LFP	Lithium iron phosphate
20	<i>Symbols</i>	
21	t	Time
22	$I(t)$	Current signal at time t
23	$U(t)$	Voltage signal at time t
24	$T(t)$	Temperature signal at time t
25	$\widehat{\text{SOC}}(t)$	Estimated state of charge at time t
26	$\widehat{\text{SOH}}(t)$	Estimated state of health at time t
27	$\text{SOC}_{\text{ref}}(t)$	Reference SOC target at time t
28	$\text{SOH}_{\text{ref}}(t)$	Reference SOH target at time t
29	ΔI	Current bias or offset
30	ΔU	Voltage bias or offset
31	ΔT	Temperature bias or offset
32	σ_I	Standard deviation of injected current noise
33	σ_U	Standard deviation of injected voltage noise
34	σ_T	Standard deviation of injected temperature noise
35	Q_c	Online cumulative charge state used by the SOC models
36	C_{nom}	Nominal capacity
37	C_{eff}	Effective capacity used inside the estimator
38	U_{max}	Upper voltage threshold used for full-charge detection
39	U_{tol}	Voltage tolerance used in the full-charge condition
40	U_1, U_2	Polarization voltages of the first and second RC branch
41	ΔMAE	Error increase relative to baseline

42 1. Introduction

43 1.1. Motivation and practical relevance

44 Reliable battery state estimation is a core BMS function because SOC and
45 SOH determine safety margins, usable energy, operating limits, charge con-
46 trol, and degradation-aware system management in both mobile and stationary
47 lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not de-
48 fined by clean-condition accuracy alone, because real BMS operation is exposed
49 to sensor bias, additive noise, initialization uncertainty after restart, missing
50 samples, irregular sampling, communication interruptions, and transient signal
51 spikes caused by measurement-chain limitations or disturbances in the surround-
52 ing system [4, 5, 6]. The relevant engineering question is therefore not only
53 whether an estimator achieves low MAE under nominal measurements, but also
54 whether it converges back to a reliable state after disturbed initialization and
55 whether it remains stable under physically plausible measurement corruption.

56 As summarized conceptually in Figure 1, the present benchmark treats
57 deployment-facing performance as a three-dimensional problem consisting of
58 nominal accuracy, convergence behavior after state inconsistency, and robust-
59 ness against disturbed measurements and signal-integrity faults, embedded within
60 broader BMS requirements such as hardware constraints and implementation
61 effort. The present study therefore isolates the performance branch and com-
62 pares representative estimator classes under identical measurement-side distur-
63 bances and a matched embedded-feasibility constraint. The benchmark shows
64 that strong nominal accuracy does not necessarily imply strong disturbance
65 performance and that the estimator classes respond differently to bias, missing
66 samples, and re-initialization events.

67 1.2. State of the art in battery state estimation

68 The literature commonly groups battery-state estimators into direct-measurement
69 or integration-based approaches, model-based approaches based on equivalent-
70 circuit or electrochemical models, hybrid approaches that combine physical

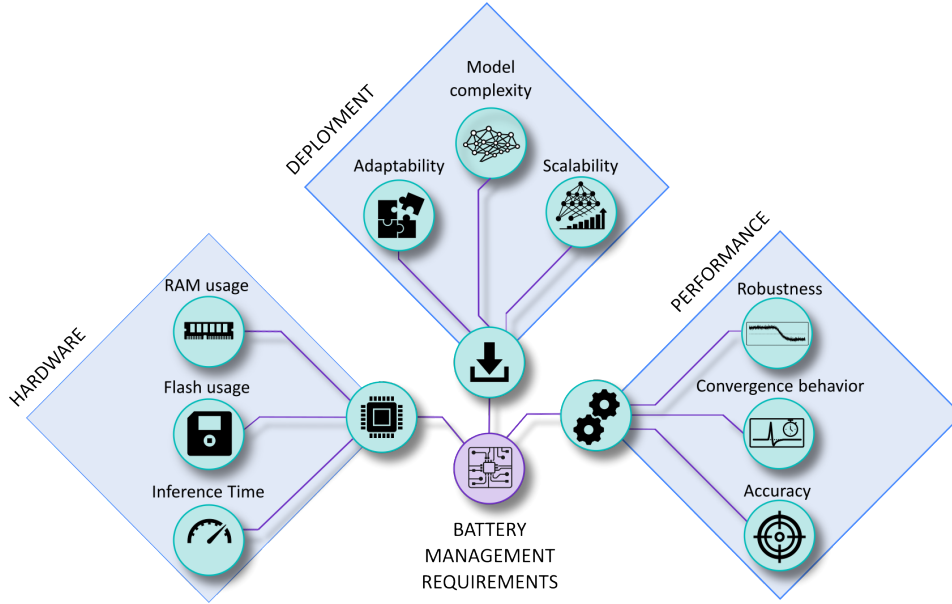


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

71 structure with learned components, and purely data-driven models based on
72 neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically
73 based on Coulomb counting, are attractive because they are simple, computa-
74 tionally inexpensive, and easy to deploy; however, they remain structurally sen-
75 sitive to current bias, missing data, and initialization errors because they lack an
76 internal correction mechanism [7, 4]. Model-based approaches, especially ECM-
77 and observer-based estimators, offer physical interpretability and voltage-based
78 correction capability, but their practical behavior depends strongly on the qual-
79 ity of model parametrization, operating-point observability, and measurement
80 quality [8, 2, 4].

81 Hybrid estimators seek to combine physical state propagation with learned
82 adaptation of capacity, parameters, or auxiliary states, which has made joint

83 SOC/SOH estimation an active research direction [9, 10, 11]. This combination
 84 can improve baseline calibration, while it does not necessarily eliminate struc-
 85 tural sensitivity to measurement disturbances. Data-driven approaches based
 86 on recurrent or sequence models can exploit nonlinear cross-relations between
 87 current, voltage, temperature, and derived online features, and recent work has
 88 also demonstrated their embedded feasibility through TinyML-oriented imple-
 89 mentations, pruning, and quantization [12, 13, 14, 15, 3].

90 *1.3. Performance and embedded deployment gap*

91 A large share of the SOC estimation literature still emphasizes clean-data ac-
 92 curacy, average error metrics, or computational efficiency, while comparatively
 93 fewer studies analyze how estimators behave under realistic measurement faults
 94 and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain
 95 confined to a single model family, for example observer-based robustness against
 96 modeling and measurement errors [4] or robust estimation concepts for specific
 97 BMS architectures under noise and uncertainty [5], whereas cross-family com-
 98 parisons under the same disturbance protocol and under comparable embedded
 99 constraints remain limited. Benchmarking studies in the broader BMS con-
 100 text underline the importance of reproducible scenario design and deployment-
 101 relevant validation, but they do not by themselves close the gap between clean-
 102 data estimator comparison and measurement-level performance benchmarking
 103 across fundamentally different estimator classes [6]. This gap is especially rele-
 104 vant in embedded BMS design, where estimator selection must jointly consider
 105 nominal accuracy, convergence, robustness under disturbed measurements, and
 106 computational constraints.

107 *1.4. Contribution and study objectives*

108 This work defines deployment-facing estimator performance as the joint as-
 109 sessment of nominal accuracy, convergence behavior after state inconsistency,
 110 and robustness under realistic disturbances applied only to measurable inputs

and their timing. The study compares four deployment-prepared representatives of major estimator families, namely a direct-measurement model (DM), a hybrid direct-measurement model with learned SOH support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven neural model with rolling-window GRU-based SOC estimation and stateful LSTM-based online SOH context (DD), under identical causal execution conditions. The disturbances are deliberately restricted to measurement-level and signal-integrity scenarios, including current bias, current noise, voltage noise, temperature noise, missing samples, irregular sampling, burst dropouts, voltage spikes, ADC quantization, and initialization errors, so that the benchmark remains physically interpretable and experimentally reproducible.

The manuscript should therefore be read as a controlled cross-class estimator benchmark and not as a complete system-level BMS validation campaign. Its role is to isolate estimator-side failure modes under matched data, matched disturbance definitions, matched online execution rules, and matched embedded-feasibility constraints. The resulting contribution is a measurement-only benchmark that separates baseline accuracy, convergence, and disturbed-operation robustness and translates the observed failure modes into decision-oriented guidance for embedded BMS estimator selection.

2. Methodology

2.1. Estimator classes under test

Figure 2 summarizes the causal benchmark chain, from measured input signals and online feature reconstruction to disturbance injection and the final global and local performance evaluation. The direct-measurement model (DM) represents the canonical embedded Coulomb-counting approach with fixed nominal capacity and explicit full-charge reset logic, and serves as the low-complexity reference for integration-driven SOC estimation. All SOH-dependent estimator variants in the benchmark, namely HDM, HECM, and DD, use the same shared LSTM-based online SOH estimator; the classes differ only in how this

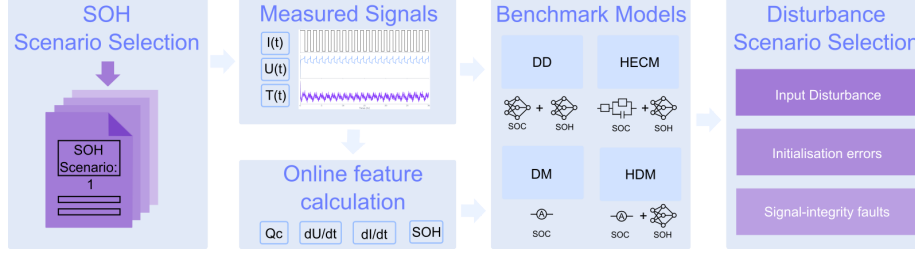


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator classes, exposed to measurement-only disturbances, and finally evaluated through global and local performance metrics.

140 common SOH signal is consumed inside the SOC pipeline. The hybrid direct-
 141 measurement model (HDM) preserves the same Coulomb-counting core but re-
 142 places the fixed capacity assumption with an online effective capacity derived
 143 from this shared SOH estimator, which makes it representative of lightweight
 144 hybrid estimation. The hybrid equivalent-circuit model (HECM) represents
 145 physics-based state estimation with observer correction, because it combines
 146 current-driven state propagation with voltage-residual feedback and SOH-dependent
 147 parameterization in a second-order RC equivalent-circuit structure driven by the
 148 same shared SOH estimate [8, 2]. The data-driven model (DD) represents a neu-
 149 ral estimator pipeline in which SOC is predicted by a lightweight rolling-window
 150 GRU sequence model while the shared LSTM-based online SOH estimate is pro-
 151 vided as causal context under the same execution constraints [12, 3].

152 2.2. SOC estimation methods

153 2.2.1. Direct-measurement model (DM)

The direct-measurement SOC formulation used in DM, HDM, and as an explicit auxiliary feature in DD follows the same online cumulative-charge state, denoted here uniformly as Q_c . It is updated causally from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

where I_k is given in ampere and Δt_k in seconds, so that the factor 3600 s h^{-1} converts the increment to ampere-hours. The sign convention is fixed by the benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right), \quad (2)$$

154 2.2.2. Hybrid direct-measurement model (HDM)

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

155 This formulation makes the relation between DM and HDM explicit: both use
 156 the same online charge state Q_c , while only the denominator changes through
 157 online SOH support. In both direct-measurement variants, a sustained near-
 158 maximum voltage condition is interpreted as a constant-voltage full-charge event;
 159 if the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible volt-
 160 age margin below the upper threshold U_{max} , for the configured CV duration, the
 161 internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$, which re-
 162 flects the practical assumption that a completed CV phase provides a physically
 163 anchored full-charge reference.

164 2.2.3. Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

where U_1 and U_2 denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta\Delta t_k/C_b \\ R_1(1 - e^{-\Delta t_k/\tau_1}) \\ R_2(1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

where $C_b = C_0\widehat{\text{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\widehat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

and the Kalman update uses the residual between $\widehat{U}_k$ and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample I_{k-1} , whereas the measurement equation uses the current sample I_k available at the correction step. The ECM parameters are not constant. The implementation interpolates R_i , R_1 , R_2 , τ_1 , τ_2 , OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$. The benchmarked structurally pruned and fine-tuned checkpoint consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length $L_{\text{SOC}} = 2024$ samples. Within this feature set, Q_c is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as finite differences with respect to the current sampling interval,

$$\frac{dI}{dt}(k) = \frac{I_k - I_{k-1}}{\Delta t_k}, \quad \frac{dU}{dt}(k) = \frac{U_k - U_{k-1}}{\Delta t_k}, \quad (9)$$

and provide the neural estimator with local transient information on current and voltage dynamics that is not directly captured by the absolute channel values alone. The additional feature Δt_k is passed explicitly so that the network can distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. No future information is used. Continuous hidden-state forwarding is evaluated only as a separate deployment ablation because it changes the learned inference semantics.

2.3. SOH estimation method

The shared SOH estimator used in the current benchmark is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{EFC}, Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information; the first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and then held piecewise constant between update points, which enforces an online deployment constraint and avoids sample-wise access to future information while still allowing the faster SOC es-

Table 1: Estimated persistent flash and RAM footprints of the reusable estimator components for the deployment-prepared variants. For the recurrent neural components, flash estimates refer to the structured-pruned, finetuned, and 8-bit quantized models. RAM* includes only architecture-intrinsic persistent numerical states; temporary buffers, framework overhead, stack usage, and system-level runtime memory are excluded.

Component	Flash [KiB]	RAM* [B]	Basis
DM core (Coulomb counting)	4.2	16	four persistent scalar states in the online update logic
SOC-GRU core	27.9	268	one recurrent hidden state with size 67 stored as float32
SOH-LSTM core	408.3	1792	two-layer LSTM with hidden and cell states of size 112 stored as float32
HECM core (2RC observer)	442.5	52	observer state $x \in \mathbb{R}^3$, covariance $P \in \mathbb{R}^{3 \times 3}$, and previous current

timators to use the latest SOH context. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

2.4. Embedded-oriented model preparation

The focus of the present paper is estimator performance and not hardware-oriented model optimization. The compared neural checkpoints use compact widths selected before the benchmark: hidden size 67 for the structurally pruned and fine-tuned SOC-GRU and hidden size 112 for the shared SOH-LSTM. The primary software benchmark evaluates these floating-point checkpoints; it does not substitute a separately quantized checkpoint. Tables 1 and 2 therefore provide analytical parameter/state-storage estimates rather than measured proof of target deployment. Actual latency, peak runtime RAM including buffers and stack, flash image size, and CPU load remain part of the separate STM32 hardware campaign.

Table 2: Estimated combined estimator footprints relative to a reference embedded budget of 1 MB flash and 1 MB RAM. The recurrent neural components correspond to the deployment-prepared variants after structured pruning, finetuning, and 8-bit quantization. RAM* reports only persistent numerical states and therefore serves as a strict lower bound rather than as a full runtime-memory measurement.

Estimator	Flash [KiB]	RAM* [B]	Flash < 1 MB	RAM* < 1 MB
DM	4.2	16	yes	yes
HDM	412.5	1808	yes	yes
HECM	850.8	1844	yes	yes
DD	436.2	2060	yes	yes

2.5. Performance benchmark design

The benchmark follows a strict measurement-only principle: only quantities that a real BMS measures or derives online from measured data are manipulated, namely current, voltage, temperature, sample availability, and sampling time, whereas artificial parameter mismatch and synthetic hidden-state corruption are deliberately excluded [6]. Figure 3 groups the benchmark into input disturbances, initialization errors, and signal-integrity faults, while Table 3 defines the exact disturbance implementations used in the study.

The disturbance magnitudes are not intended as one-to-one copies of a single field specification. Instead, they were selected from three justification classes: deployment-level sensing limitations, adverse measurement corruption, and deliberate signal-integrity stress tests [20, 21, 22, 23, 24, 25]. The main benchmark matrix follows the scenario families shown in Figure 3 and provides the basis of the main result figures and tables. ADC quantization is part of the same scenario set and is included in the cross-scenario overview; its detailed numerical results are reported in Appendix Table B.9 and Figure B.21. Overall, the benchmark is designed as a controlled and reproducible scenario study that exposes estimator-specific failure modes under a common disturbance protocol rather than emulating the complete system stack of a production BMS.

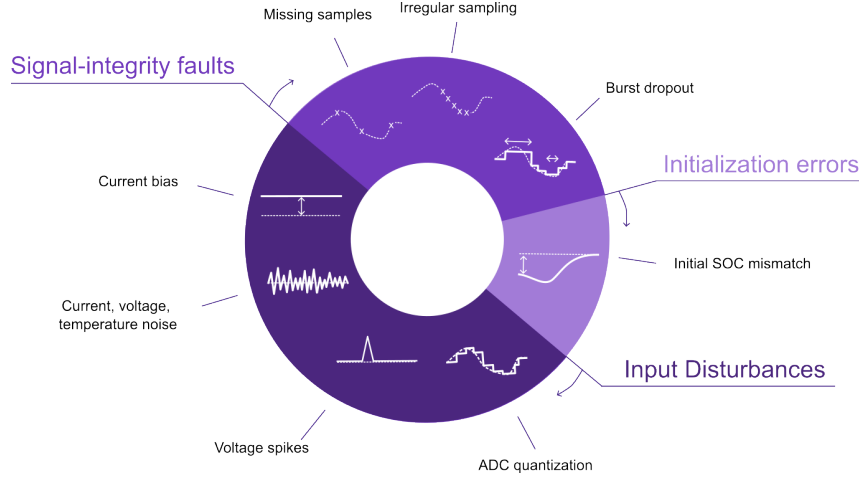


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

Table 3: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings; the cited sources motivate disturbance type and magnitude range.

Scenario	Benchmark setup	Practical origin / rationale
ADC quantization	$\Delta I = 0.01 \text{ A}$; $\Delta U = 0.005 \text{ V}$; $\Delta T = 0.5 \text{ }^\circ\text{C}$	ADC steps; sensor scaling; rounding; coarse deployment-level discretization [20, 21, 26]
Current gain error	Signed gain sweep: $I_{\text{meas}} = I(1 + b)$ with $b = \pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$	Sensor gain error; miscalibration; drift; upper bound as stress case [21, 22, 23]
Current noise	Gaussian; $\sigma_I = 0.02 \text{ A}$ and 0.10 A	Sensing-chain noise; EMI pickup; low and high sensor-noise levels [4, 5]
Voltage noise	Gaussian; $\sigma_U = 0.01 \text{ V}$	Wiring noise; front-end noise; poor filtering; EMI [4, 20, 24]
Temperature noise	Gaussian; $\sigma_T = 1.0 \text{ }^\circ\text{C}$	Sensor tolerance; conversion noise; thermal gradients; sensor-to-cell mismatch [26, 11]
Initial SOC error	Initial mismatch: -0.10 SOC ; DD via perturbed online Q_c	Restart; storage; missing recent history; weak LFP OCV anchoring [4, 8]
Missing samples	Every 50th sample frozen; separate random 2% loss	Packet loss; blocked updates; logger dropouts; periodic and random data freeze [5, 25]
Irregular sampling	Uniform jitter: $\pm 0.1 \text{ s}$, $\pm 0.5 \text{ s}$, and $\pm 0.9 \text{ s}$ around nominal 1 s grid	Scheduling jitter; bus latency; asynchronous acquisition [6, 5, 25]

Table 3: Disturbance configurations used in the benchmark and practical rationale (continued).

Scenario	Benchmark setup	Practical origin / rationale
Burst dropout	One central frozen gap: 3600 s	Communication interruption; frozen task; stuck data stream; severe recovery case [6, 5]
Voltage spikes	Single-sample spikes: ± 0.20 V every 1000 samples (1000 s)	EMI; switching disturbance; corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, convergence behavior after initialization mismatch or interrupted signal continuity, and robustness under sustained or transient measurement corruption. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the selected evaluation windows. Local disturbance performance is analyzed through scenario-specific metrics such as jump counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as window-wide MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth, recovery time, and residual post-disturbance error.

ADC quantization, current bias, voltage noise, temperature noise, missing samples, and irregular sampling are interpreted primarily through global metrics, whereas burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally use dedicated local evidence because their most relevant effects are episodic or recovery-related. Recovery comparisons use one estimator-independent criterion: absolute SOC error below 0.02 for a continuous 300 s, evaluated up to a common 24 h censoring horizon. Runs that do

not recover are retained as censored observations rather than silently omitted. The comparison reports recovery-or-censoring time, the censored fraction, initial post-event error, MAE after 1 h and 6 h, and excess-error area under the curve. The requested -0.10 initialization mismatch is mapped into each estimator’s available online state: the coulomb-counting SOC initialization for DM and HDM, the EKF SOC state for HECM, and the capacity-equivalent Q_c offset for DD. Because the DD mapping passes through the recurrent network rather than directly fixing its output, the realized initial output error is reported for every model before recovery is compared.

The independent statistical unit is the holdout cell, not an individual time sample or selected window. Predeclared SOH windows are first averaged within each cell and random seed; the six cells then receive equal macro weight. Hierarchical bootstrap resampling with seeds nested inside cells provides 95% confidence intervals. Scenario effects are evaluated as paired, cell-matched differences from baseline, and estimator pairs are compared using exact two-sided sign-flip tests, raw mean and median differences, paired standardized effect sizes, and Holm-adjusted p -values. Figure 4 summarizes this analysis chain.

3. Experimental Setup

3.1. Dataset and data acquisition

The benchmark is based on the MGFarm LFP 18650 dataset, which was generated in a laboratory environment as a controlled design-of-experiments study on cyclic aging and operating conditions and combines long full-cell trajectories with high-resolution measured signals and repeated checkup structures [27, 28, 29]. Model development and final testing use a fixed cell-disjoint split. SOC and SOH training use C01, C03, C05, C11, C17, and C23; validation uses C07, C19, and C21; and the independent holdout benchmark uses C09, C13, C15, C25, C27, and C29. The six test cells span distinct aging, thermal, duration, and load envelopes, as summarized in Figure 5. Conclusions are therefore

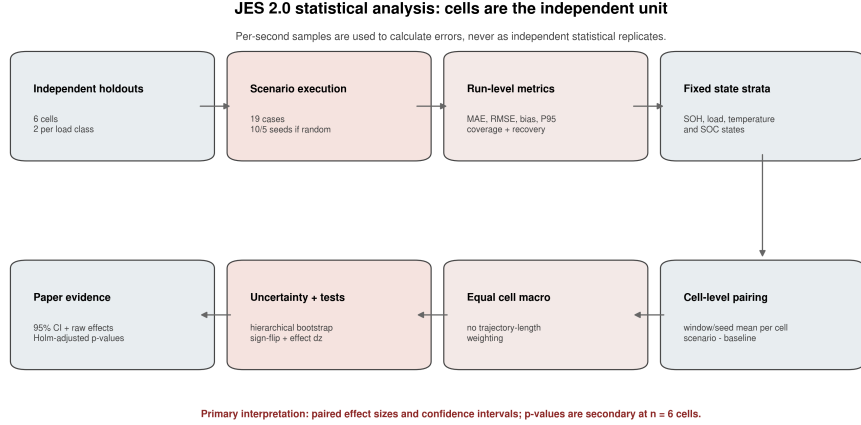


Figure 4: Statistical analysis workflow used for JES2. Measurement samples form the windows from which run-level errors are calculated, whereas cells remain the independent units for uncertainty estimation and paired inference. Windows are averaged within cell and seed, random seeds are nested within cells, and final reporting emphasizes paired effects and confidence intervals.

293 limited to the available LFP chemistry and measured operating envelope; the
 294 study does not claim cross-chemistry or pack-level generalization.

295 The holdout cells are assigned to balanced descriptive load tertiles before
 296 model results are inspected: C25 and C27 form the low-load class, C09 and
 297 C13 the middle-load class, and C15 and C29 the high-load class. Within
 298 each trajectory, performance is additionally stratified by reference SOH (fresh:
 299 $\text{SOH} \geq 0.90$; mid-life: $0.80 \leq \text{SOH} < 0.90$; aged: $\text{SOH} < 0.80$), instantaneous
 300 absolute C-rate (< 0.5 , $0.5\text{--}1.5$, and ≥ 1.5), temperature ($\leq 30^\circ\text{C}$, $30\text{--}35^\circ\text{C}$,
 301 and $> 35^\circ\text{C}$), and SOC (< 0.20 , $0.20\text{--}0.80$, and > 0.80). Figures 6–8 make the
 302 resulting coverage and missing combinations explicit; absent states are reported
 303 as not available and are never imputed.

304 3.2. Reference targets and preprocessing

305 The reference preprocessing first computes the signed transferred charge as
 306 $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the

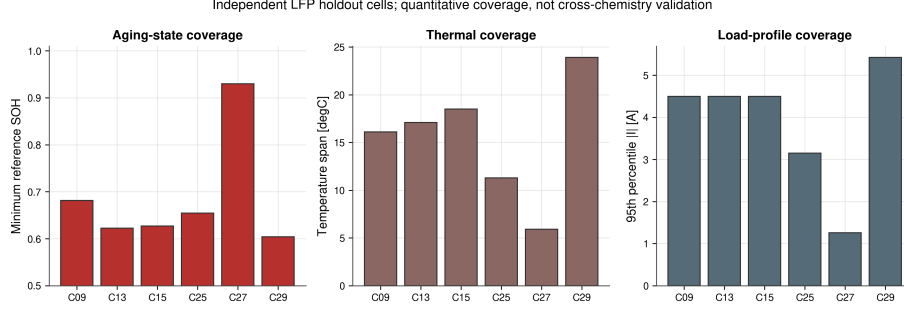


Figure 5: Quantitative coverage of the six independent holdout cells. Panel (a) shows the observed SOH range and mean, panel (b) the frozen load-class assignment based on the measured 95th-percentile absolute C-rate, panel (c) the measured temperature range and mean, and panel (d) the fraction of each full trajectory in the predeclared fresh, mid-life, and aged SOH states. The load classes are descriptive strata fixed independently of estimator errors.

307 capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$.
 308 The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}}/C_{\text{ref},0}$, where
 309 $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the
 310 resulting capacity and SOH columns are interpolated between the dedicated
 311 capacity-test anchors. The reference SOC target used in the benchmark corre-
 312 sponds to the offline Coulomb-counting target $\text{SOC}_{\text{ref}} = 1 + Q_c/C_{\text{ref},0}$, where
 313 the cumulative charge state Q_c is propagated from the signed charge and reset
 314 to 0 at the upper voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor dur-
 315 ing preprocessing. These reference targets are intentionally treated as offline
 316 evaluation quantities, because they rely on dataset-side processing and anchor
 317 logic that are not fully available to an online estimator during deployment.

3.3. Test trajectory and benchmark scenarios

319 All estimators are evaluated on every available holdout-cell/SOH combina-
 320 tion under the same disturbance protocol. One representative 24-h window is
 321 frozen for each available fresh, mid-life, and aged state, yielding 16 windows;
 322 C27 contributes only a fresh window because its measured SOH does not fall
 323 below 0.90. Windows start at a measured full-charge anchor ($\text{SOC} \geq 0.98$,

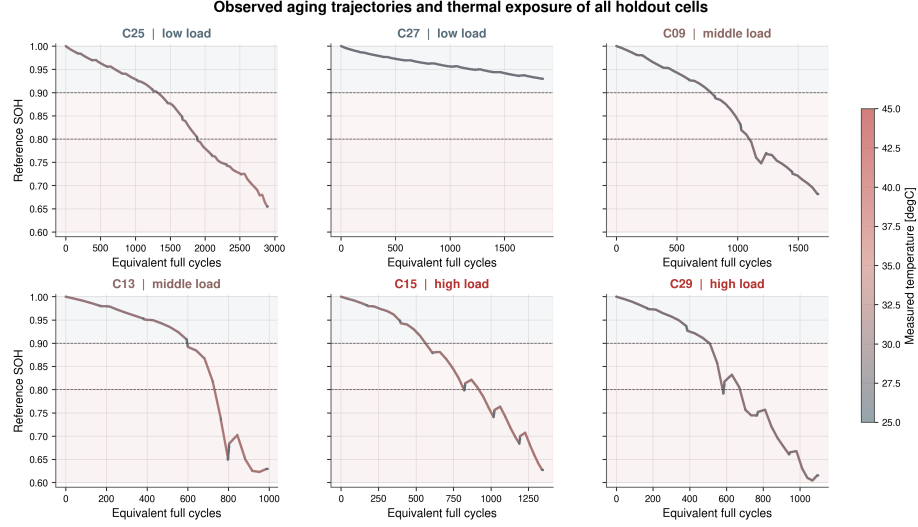


Figure 6: Observed reference-SOH trajectories over equivalent full cycles for all six holdout cells. Point color represents measured temperature. C27 provides a mild-aging control, whereas the remaining cells cover mid-life and aged operation with different degradation trajectories.

324 $U \geq 3.58 \text{ V}$) and are selected as measured-feature medoids without using es-
325 timator outputs or errors. The one-hour dropout uses 48 h from the same
326 anchor to retain approximately 24 h of post-gap observation. The shared SOH
327 LSTM receives 192 h of undisturbed causal context before every scored window,
328 matching its trained sequence length. Figure 9 summarizes this protocol.

329 The 19 predeclared cases include the nominal baseline, two current-noise
330 levels, voltage and temperature noise, three current-bias levels, voltage and
331 temperature offsets, ADC quantization, initial SOC mismatch, periodic and
332 random sample loss, three timing-jitter levels, a one-hour burst dropout, and
333 randomized-sign voltage spikes. Gaussian sensor-noise cases use 10 determinis-
334 tic seeds; random sample loss, jitter, and spike cases use 5 seeds; deterministic
335 cases run once. Figure 10 maps each case to the manipulated measurement
336 or state and marks the selected paired reference-SOH ablations. Input distur-
337 bances are applied directly to measured channels, initialization errors only where

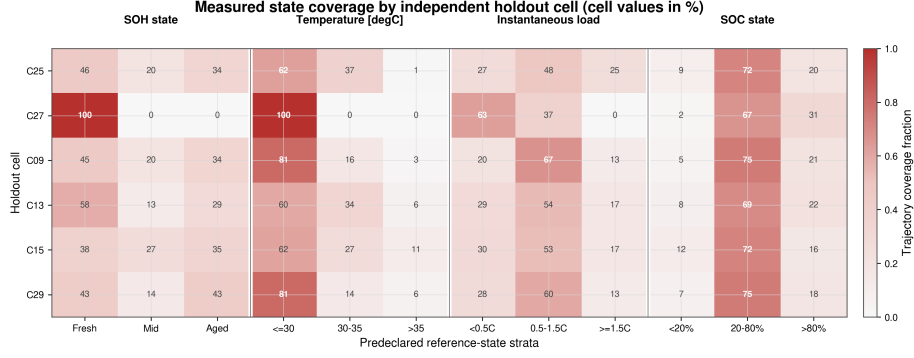


Figure 7: Fraction of each complete holdout trajectory in the predeclared SOH, temperature, instantaneous-load, and SOC strata. Values are percentages calculated from the complete trajectories, not from plotting samples. The matrix exposes unsupported combinations, including the absence of mid-life and aged states for C27.

structurally equivalent, and signal-integrity scenarios to sample availability or timing while preserving causal estimator execution.

3.4. Implementation details

Every estimator is executed causally, and all auxiliary quantities required by the models are rebuilt directly from disturbed inputs. This applies to Q_c , EFC, dU/dt , dI/dt , and hourly SOH aggregates, so no hidden dataset-side feature leaks into the estimator input. The SOH LSTM forwards hidden and cell states across hourly updates after its 192-h initialization context. The primary SOC-GRU evaluation preserves its trained sequence-to-one semantics: each prediction uses the latest 2024-sample rolling window with recurrent state reset between windows. A faster continuous-stateful stream is treated only as a deployment ablation because it changed full-trajectory DD accuracy and therefore cannot silently replace the trained benchmark semantics. In freeze and dropout scenarios, disturbed or frozen measurements propagate consistently into derived online features. Direct-measurement estimators use the same CV-triggered reset that defines the full-charge start, whereas the DD initial-SOC test uses an equivalent offset of online Q_c because the network exposes no explicit SOC state. The en-

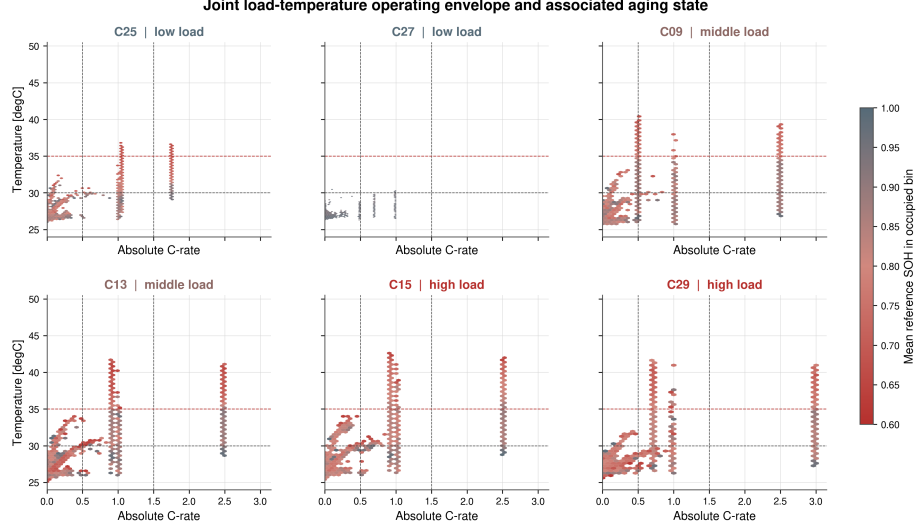


Figure 8: Joint absolute-C-rate and temperature operating envelopes of the six holdout cells. Color denotes the mean reference SOH in each occupied bin. Dashed lines show the fixed instantaneous-load and temperature-state boundaries used for stratified performance reporting.

355 vironment enforces identical capacity assumptions, disturbed streams, targets,
 356 and metrics across all estimators.

357 4. Results

358 4.1. Baseline performance

359 Figure 11 summarizes the nominal baseline performance on the selected
 360 benchmark trajectory. The hybrid direct-measurement model (HDM) achieves
 361 the lowest baseline error with $\text{MAE} = 0.0024$, followed by the hybrid equivalent-
 362 circuit model with 0.0090, the data-driven model with 0.0100, and the direct-
 363 measurement model with 0.0311, making HDM the clean-condition leader. This
 364 baseline ranking serves as the reference point for the following disturbed-measurement
 365 scenarios and for the corresponding robustness comparisons; the complete base-
 366 line metrics are listed in Appendix Table A.5.

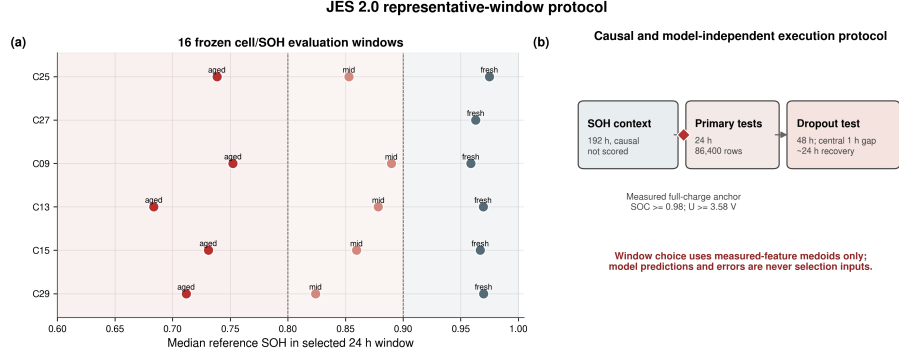


Figure 9: Frozen representative-window protocol. Panel (a) shows the 16 measured cell/SOH windows; absent states are not imputed. Panel (b) separates the unscored 192-h causal SOH context, the 24-h primary evaluation, and the 48-h dropout evaluation. Selection uses measured operating features only and is independent of estimator errors.

4.2. Current-bias sensitivity

Figure 12 summarizes the adverse response to a signed current-gain sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. For each magnitude and holdout cell, the larger MAE change from the two signs is reported because the opposite sign can partially compensate a pre-existing baseline error. The resulting worst-case degradation increases monotonically with gain-error magnitude for every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is least sensitive. At 3%, the cell-macro MAE increases by 0.0107 for DM, 0.0100 for HDM, 0.0059 for HECM, and 0.0038 for DD.

4.3. Noise sensitivity

Current noise is weaker than current bias in global MAE, but it remains informative through local behavior and output variability. In the compact matrix, the reported high-noise case uses $\sigma_I = 0.10$ A, while the local detail sweep in Figure 13 extends the current-noise axis up to $\sigma_I = 0.20$ A. Under the high-noise case, the hybrid equivalent-circuit model shows the largest global penalty ($\Delta\text{MAE} = 0.0075$), whereas DM, HDM, and DD remain close to their baseline MAE values. The local noise story differs from the global metric: the data-

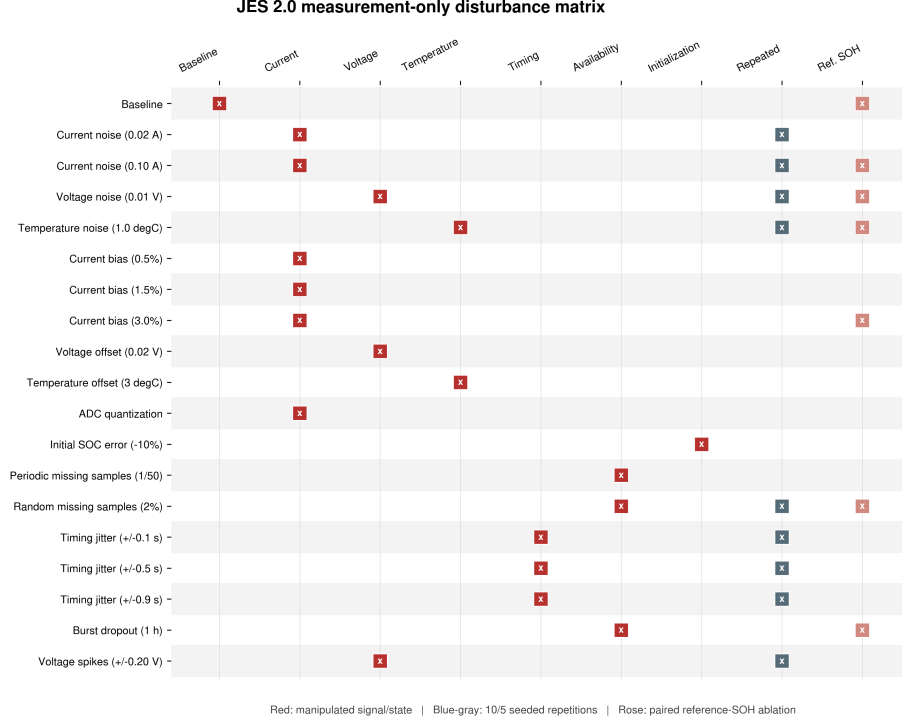


Figure 10: Complete JES2 measurement-only test matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel; blue-gray markers denote stochastic scenarios evaluated with the predeclared 10- or 5-seed budget across all available windows; rose markers identify selected paired reference-SOH ablations.

driven model exhibits the strongest output variability because current noise propagates into several derived input channels, whereas the hybrid equivalent-circuit model is the most sensitive in the error domain. The broader noise block is not limited to current noise: temperature noise with $\sigma_T = 1.0^\circ\text{C}$ is almost neutral for DM and HECM but increases the error of HDM and DD, while voltage noise with $\sigma_U = 0.01\text{ V}$ is one of the strongest discriminators because HECM remains nearly unaffected whereas DM, HDM, and DD degrade strongly. Figure 13 therefore separates the applied current disturbance, a matched SOC comparison in the same window, the global MAE penalty, and the local output-variability summary, while Appendix Tables A.6 and A.7 provide

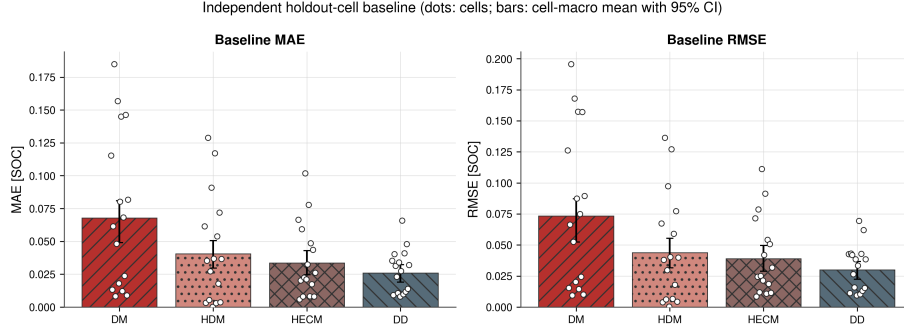


Figure 11: Baseline performance on the selected representative holdout-cell windows. The hybrid direct-measurement model provides the lowest clean-condition error, whereas the direct-measurement model shows the largest baseline deviation.

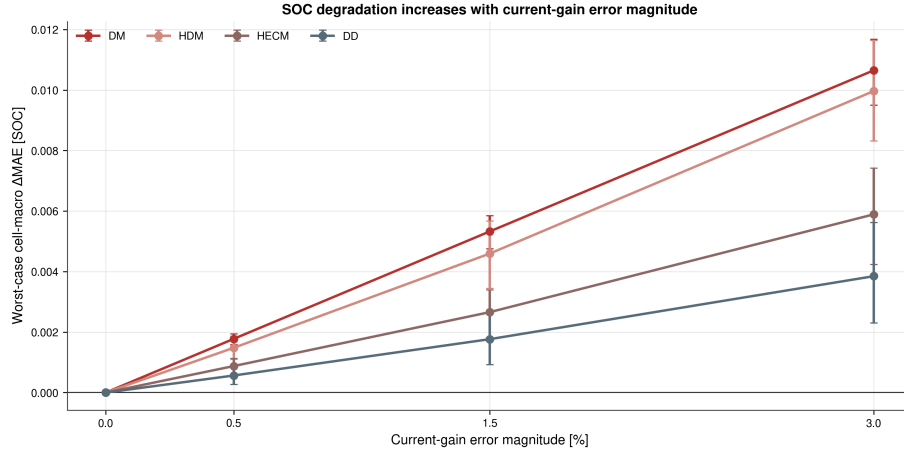


Figure 12: Worst-case sensitivity to current-gain error across the six independent holdout cells. For each gain-error magnitude and cell, the adverse direction from the matched $\pm b$ pair is retained. Points show equal-weight cell-macro changes in MAE relative to each estimator's baseline; error bars show 95% bootstrap confidence intervals across cells.

394 the complementary numerical evidence for current, voltage, and temperature
 395 noise.

396 4.4. Initialization error and recovery

397 Initialization performance must be analyzed locally and in an evaluation
 398 window without constant-voltage reset events, because global MAE hides the

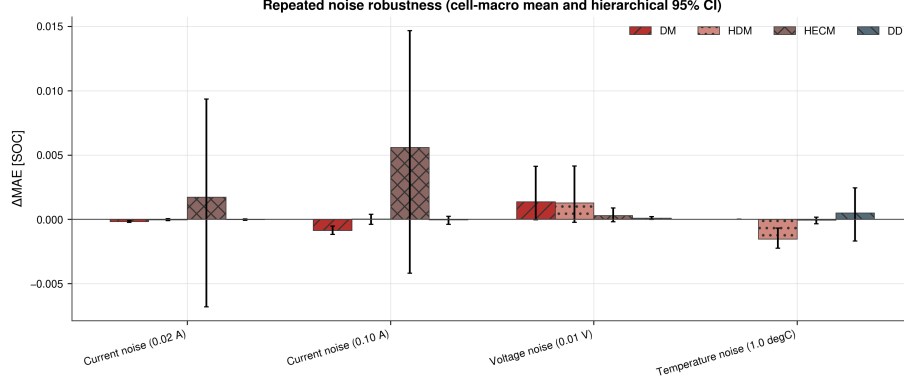


Figure 13: Noise sensitivity under additive current noise. The figure combines an applied current-noise example, a matched SOC comparison in the same 30 min window, the global MAE penalty, and the local output-variability sensitivity across noise levels.

399 actual re-synchronization behavior after a wrong initial SOC. The injected mis-
400 match in the benchmark is -0.10 SOC, as summarized in Table 3. DM recovers
401 to the strict baseline band after about 1.17 h, HECM recovers after about
402 2.37 h under the strict criterion and after about 1.16 h under the fair crite-
403 rion, whereas HDM does not re-enter either band in the inspected window. The
404 data-driven model is evaluated through an initial perturbation of the online Q_c
405 feature rather than through an explicit hard SOC-state reset, which keeps the
406 test aligned with deployment-time observables; under this formulation it recov-
407 ers after about 1.05 h under the strict criterion and after about 1.00 h under
408 the fair criterion. Figure 14 visualizes the resulting reset-free recovery behavior,
409 and the corresponding recovery metrics and baseline-band thresholds are listed
410 in Appendix Table A.7.

411 4.5. Missing samples and signal integrity

412 Missing samples are the strongest global signal-integrity stressor in the bench-
413 mark. In this scenario, every 50th sample is frozen, corresponding to one held
414 sample every 50 s on the nominal 1 s grid; under this protocol the data-driven
415 model changes only marginally from 0.0100 to 0.0117 MAE, whereas DM and
416 HDM degrade much more strongly. The irregular-sampling scenario jitters the

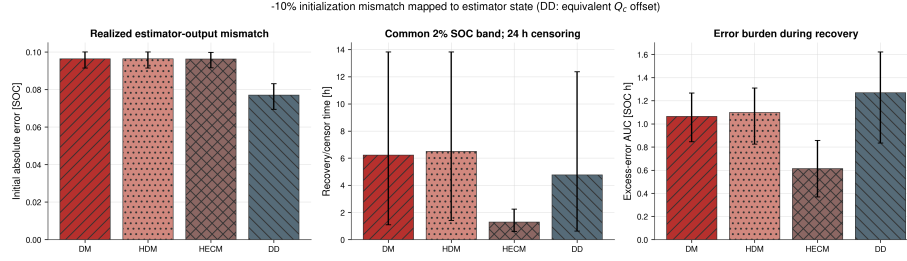


Figure 14: Initial-state recovery in a reset-free evaluation window. The top panel shows SOC trajectories after the imposed state mismatch, and the lower panel shows the corresponding absolute SOC error over time.

417 sampling intervals uniformly by ± 0.1 s around the nominal 1 s grid while pre-
 418 serving the full-run duration. Under this resampled implementation it remains
 419 nearly neutral for all models, with only very small signed MAE changes around
 420 zero. The burst-dropout scenario consists of one central frozen gap of 3600 s. It
 421 remains globally moderate, but all models require roughly 27–29 h to return to
 422 the baseline error band, making the post-gap recovery behavior more informa-
 423 tive than the global MAE alone. The signal-integrity block therefore needs both
 424 global and local views: Figure 15 summarizes the global penalties for missing
 425 samples, irregular sampling, and burst dropout, Figure 16 shows the transition
 426 into the gap, and Figure 17 shows the long recovery afterwards. The corre-
 427 sponding global metrics are listed in Appendix Table A.6, the burst-dropout
 428 recovery times in Appendix Table A.7, and the scenario-to-evidence mapping
 429 in Appendix Table A.8.

430 4.6. Spike sensitivity

431 Single voltage spikes have very different local effects across model classes,
 432 even when the full-run MAE changes remain small for DM, HDM, and HECM.
 433 The spike protocol injects one ± 0.20 V voltage spike every 1000 samples, corre-
 434 sponding to every 1000 s on the nominal 1 s time base. The data-driven model
 435 shows the clearest spike sensitivity, both in the representative SOC window and
 436 in the local aligned error response, with a noticeable global increase from 0.0100

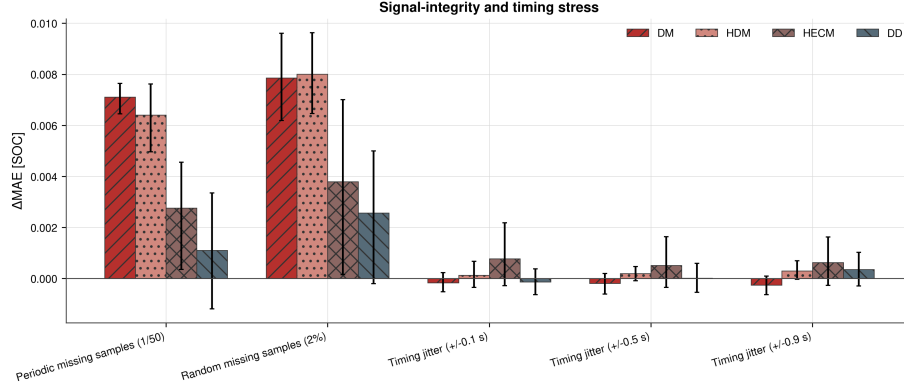


Figure 15: Signal-integrity performance summary. The left panel compares the global MAE penalties for missing samples, irregular sampling, and burst dropout, whereas the right panel summarizes the burst-dropout recovery time.

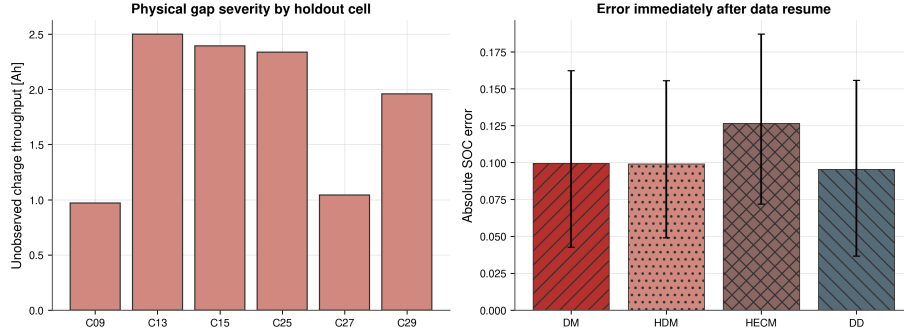


Figure 16: Local transition into the burst-dropout window. The shaded region marks the missing-data interval, and the lower panel shows how the estimators behave while the measurements are frozen.

437 to 0.0144 MAE in the present spike protocol. Figure 18 summarizes this behav-
 438 ior through a representative local window, a spike-severity summary, and the
 439 aligned post-spike error response. To avoid misleading signed full-run averages,
 440 the compact spike summary is reported as the 95th percentile of the post-spike
 441 peak excess error rather than as signed Δ MAE; the numerical spike-related
 442 metrics are summarized in Appendix Tables A.6 and A.7, and the role of local
 443 evidence for spikes is highlighted in Appendix Table A.8.

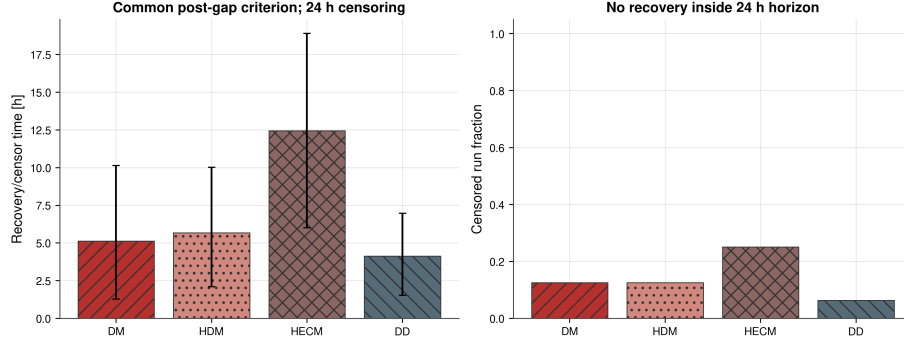


Figure 17: Post-gap recovery after the burst dropout. The upper panel shows the SOC trajectories after the gap, and the lower panel shows the corresponding absolute SOC error together with the recovery markers.

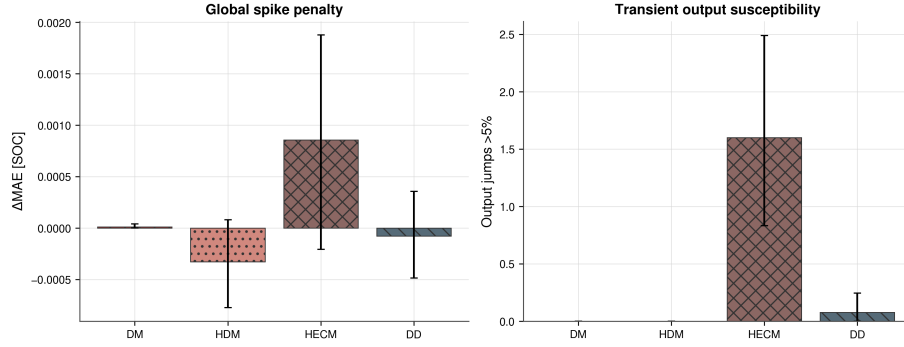


Figure 18: Voltage-spike sensitivity. The figure combines a representative SOC window around a spike, a local spike-severity summary based on the 95th percentile of the post-spike peak excess error, and the aligned local excess-error response across repeated spikes.

4.7. Cross-scenario comparison

Figure 19 condenses the disturbed-scenario penalties into one cross-scenario view and makes the estimator-specific pattern shifts directly visible; it also includes the ADC-quantization case. The cross-scenario view confirms that the benchmark does not yield one universal winner across all criteria, because HDM is best under nominal conditions, HECM is most resilient under current bias and voltage noise, DD is the most stable estimator under missing samples, and DD also shows the fastest strict-band recovery after initialization mismatch.

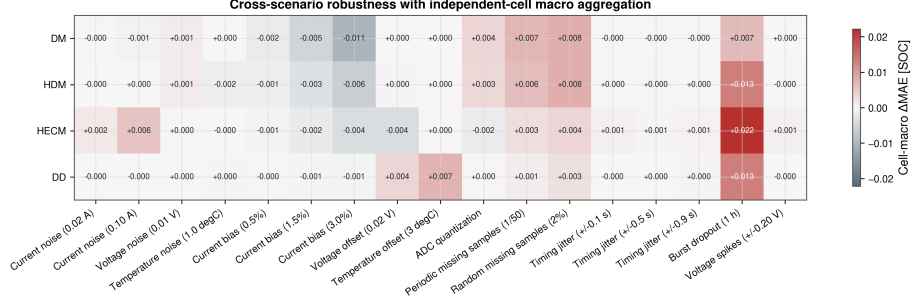


Figure 19: Cross-scenario error increase relative to the baseline case. The heatmap summarizes the disturbed-scenario ΔMAE values, including ADC quantization, and highlights where robustness rankings change across disturbance classes.

Current bias and voltage noise are the most discriminative scenarios in the present benchmark phase because they most clearly separate correction-capable estimators from purely integration-driven or fully data-driven alternatives. The heatmap is useful because it shows where the ranking changes across disturbance classes and where estimator strengths are scenario-specific rather than universal; the corresponding numerical overview is compiled in Appendix Tables A.5–A.7.

5. Discussion

5.1. Model-specific performance behavior

The direct-measurement model acts as the canonical drift-prone estimator, because every persistent current disturbance directly accumulates in the SOC estimate and missing measurement information cannot be corrected by an additional feedback mechanism. The hybrid direct-measurement model improves nominal calibration through online capacity adaptation, but it remains structurally exposed to the same current-integration failure mode and additionally inherits the dynamics of the learned SOH support. The hybrid equivalent-circuit model, combined here with the same shared data-driven online SOH estimator used by the other SOH-aware variants, benefits from interpretable ECM parameterization, explicit voltage-residual correction, and causal observer structure.

470 This combination explains why it handles current bias and voltage noise much
471 better than the integration-based alternatives, even though it becomes more
472 sensitive when noise perturbs the current-dependent parameter selection and
473 observer update.

474 The present data-driven neural model behaves as a compact context-rich
475 estimator whose robustness depends strongly on the disturbance channel: it
476 is notably stable under missing samples, competitive under current bias, and
477 the fastest model after initialization mismatch, but more exposed to derivative-
478 amplified noise and voltage spikes. Read together, these results suggest that
479 the current compact instance still relies too strongly on dominant trigger fea-
480 tures from the input stream and does not yet exploit voltage and temperature
481 information as effectively as the best hybrid physics-based alternative. That
482 observation should be read as a design lesson rather than as a class limitation.
483 Unlike the fixed-form direct-measurement and ECM estimators, the data-driven
484 class retains a comparatively large tuning space through architecture, retraining,
485 feature selection, and deployment-time state handling, so its present benchmark
486 ranking reflects one compact embedded instance rather than a hard ceiling of
487 the class itself.

488 5.2. Accuracy–performance trade-offs

489 The benchmark makes clear that low baseline MAE does not imply robust
490 disturbed operation, because the best nominal model under clean measurements
491 is not the best model under current bias or voltage noise. The observed trade-
492 off is therefore not only between accuracy and computational complexity, but
493 also between calibration strength, explicit physical correction capability, re-
494 covery speed after state inconsistency, and sensitivity to disturbance propaga-
495 tion through learned or derived features. The benchmark consequently yields
496 a structured performance split: the hybrid direct-measurement model is the
497 nominal-accuracy leader, the hybrid equivalent-circuit model is the overall ro-
498 bustness leader, and the data-driven neural model converges back to the strict
499 baseline band fastest after initialization mismatch. This result argues against

single-metric leaderboard thinking and favors a scenario-aware interpretation in which estimator quality is tied to the disturbance classes and recovery requirements that matter most in the target application.

5.3. Implications for BMS deployment

If current-sensor bias and long-term drift are dominant concerns, the benchmark evidence favors the hybrid equivalent-circuit approach, because the observer can re-anchor the SOC estimate through the measured voltage while the shared online SOH model keeps the parameterization aging-aware. If the deployment focus is nominal accuracy under controlled sensing and low-compute execution, the hybrid direct-measurement class remains attractive, but its vulnerability to biased current measurements must be considered explicitly. If rapid recovery after restart or initial-state mismatch is the primary requirement, the data-driven neural approach is attractive because it re-enters the strict baseline band fastest in the present benchmark, while simultaneously remaining strong under missing-sample stress. If robustness against signal-integrity issues such as missing samples is important, the data-driven model offers a strong alternative, provided that the input feature pipeline is designed so that the estimator does not over-focus on a single dominant measurement channel and remains stable under voltage-driven disturbances.

While the present study focuses on performance, practical BMS deployment also requires consideration of implementation complexity and scalability when the observed estimator behavior is transferred to real systems. Within that broader design space, the data-driven class retains the clearest improvement headroom because architecture, features, and training can be updated directly, whereas ECM-based estimators are more constrained by model topology and parameter-identification workflow. At the same time, the hybrid equivalent-circuit configuration used here shows why a physics-based SOC core plus shared data-driven SOH support can already yield a very strong compromise between interpretability, disturbance correction, and embedded feasibility. The footprint estimates indicate that persistent state memory is negligible for all estimators

530 in the present study and that all four deployment-prepared estimator variants
 531 remain below the reference 1 MB flash budget. The tightest flash case is the hy-
 532 brid equivalent-circuit estimator because it combines the ECM lookup structure
 533 with the shared SOH-LSTM, whereas the direct-measurement baseline keeps
 534 the largest margin. For embedded BMS design, the practical selection criterion
 535 should therefore combine the expected fault profile of the measurement chain
 536 with implementation constraints. Table 4 condenses the benchmark outcomes
 537 into a decision-oriented recommendation map so that the reader is not left with
 538 a large set of disconnected scenario results.

539 Figure 20 complements this recommendation map with an illustrative vi-
 540 sual synthesis of the same benchmark evidence. The values in this figure
 541 are relative scores rather than additional physical measurements. For each
 542 lower-is-better metric, the four estimator classes are min-max normalized as
 543 $(x_{\max} - x)/(x_{\max} - x_{\min})$, so that the best class in the present comparison re-
 544 ceives 1 and the weakest class receives 0. The accuracy dimension averages the
 545 normalized baseline MAE, RMSE, and 95th-percentile error. The robustness di-
 546 mension averages the normalized Δ MAE values across the disturbed scenarios
 547 listed in Appendix Table A.8. The recovery dimension uses the strict recovery
 548 time after the initial-SOC perturbation; missing recovery within the inspected
 549 window is treated as non-recovery and assigned the lowest score. The bar chart
 550 then applies three illustrative priority weightings to these three dimensions:
 551 accuracy-weighted (0.60/0.20/0.20), robustness-weighted (0.20/0.60/0.20), and
 552 recovery-weighted (0.20/0.20/0.60), where the entries correspond to accuracy,
 553 robustness, and recovery, respectively. Thus, each profile assigns 60% of the
 554 composite score to its focal dimension and 20% to each of the other two dimen-
 555 sions. The accuracy-weighted profile should therefore be read as an accuracy-
 556 prioritized compromise, not as the pure baseline-accuracy ranking reported in
 557 Table 4. The synthesis is intended as a compact orientation aid and does not
 558 replace the raw scenario-wise results.

Table 4: Decision-oriented recommendation map derived from the benchmark results. The table summarizes which estimator class is most suitable when one deployment priority dominates the design decision.

Primary deployment priority	Recommended class	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error under clean sensing	HDM	Lowest baseline MAE on the benchmark trajectory (0.0024)	Strong sensitivity to current bias and no recovery to the baseline bands in the inspected initialization-mismatch window
Best overall robustness under uncertain measurement quality	HECM	Strongest current-bias robustness, nearly unchanged under voltage noise, and competitive baseline accuracy	Larger flash footprint and slower strict-band recovery after initialization mismatch than DD
Fastest re-synchronization after restart or wrong initial state	DD	Fastest strict-band recovery in the local initialization analysis	More sensitive than HECM to voltage spikes and less robust than HECM under current bias
Highest resilience to missing samples and signal freezing	DD	Smallest MAE increase under missing samples and strong post-disturbance stability	Clean-condition accuracy remains below HDM, and voltage-driven disturbances require careful feature-pipeline design
Best development headroom for iterative model improvement	DD	Architecture, features, and training can be updated directly while retaining embedded-feasible deployment and strong performance in recovery and missing-sample scenarios	Present compact instance is still weaker than HECM in the strongest bias- and voltage-driven stress cases
Simplest implementation and largest resource margin	DM	Minimal architecture and smallest flash footprint by a wide margin	Weakest robustness under biased current measurements and missing-sample stress

5.4. Limitations and future extensions

The current benchmark phase is based on one representative dataset trajectory and should later be expanded to additional cells, broader aging states, and cross-cell validation before stronger generalization claims are made. The spike study currently uses isolated spike events; clustered disturbances, longer spike sequences, and hardware-level disturbance injection may reveal additional model differences. The initialization test for the data-driven model remains an online-state proxy based on Q_c perturbation rather than an explicit latent-state reset, which is methodologically justified but should still be interpreted as a class-specific approximation. Future extensions should include the remaining extended scenarios, additional cells, pack-level settings, and eventually

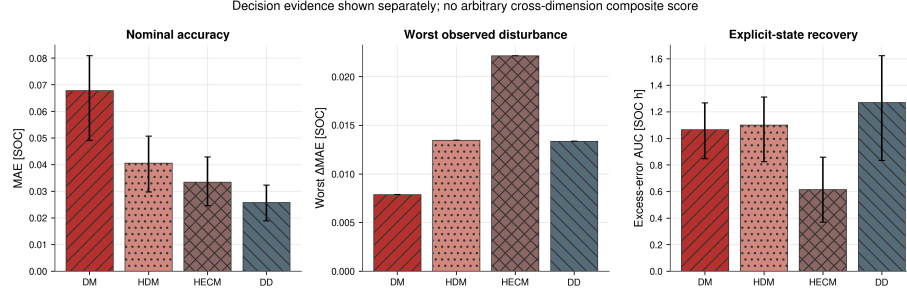


Figure 20: Decision-oriented synthesis of the benchmark results. Panel (a) collapses the benchmark into three relative, min-max-normalized decision dimensions, namely nominal accuracy, disturbed-scenario robustness, and recovery behavior. Panel (b) shows three illustrative deployment-priority profiles and is intended only as a visual complement to Table 4, not as a replacement for the raw scenario metrics.

hardware-in-the-loop or true embedded runtime benchmarks under the same performance-oriented evaluation philosophy.

6. Conclusion

This work frames battery-state estimator evaluation as a performance problem under realistic measurement corruption rather than as a clean-data accuracy comparison alone. The central engineering message is that estimator selection depends on the expected disturbance profile of the BMS measurement chain, the required recovery behavior after corrupted inputs, and the practical implementation constraints of the target system, because no single estimator dominates all deployment-relevant dimensions. The main findings can be summarized by estimator family: the hybrid direct-measurement model is the nominal-accuracy leader, the hybrid equivalent-circuit model with shared data-driven SOH support is the strongest overall robustness choice in the most discriminative disturbance scenarios, and the data-driven neural model is strongest in recovery after initialization mismatch and in missing-sample resilience.

If one general-purpose estimator must be preferred under the specific disturbance profile emphasized here, the benchmark supports the hybrid equivalent-

587 circuit approach as the strongest overall compromise. Its combination of in-
 588 terpretable physical parameterization, voltage-based correction, and compact
 589 shared SOH support preserves a clear role for ECM-based estimation in embed-
 590 ded BMS design. At the same time, the data-driven model class remains highly
 591 attractive when the development strategy includes iterative data collection and
 592 model updates. The present benchmark also yields a practical design lesson:
 593 data-driven SOC estimators should not rely too strongly on one dominant trig-
 594 ger variable, but should be trained and feature-engineered to make balanced
 595 use of the available current, voltage, temperature, and health context if they
 596 are to realize their full robustness potential. The benchmark is designed to be
 597 extensible, and the next steps are the extension to more cells and scenarios, the
 598 inclusion of remaining disturbance classes, and the transfer of the methodology
 599 toward broader embedded or hardware-in-the-loop validation.

600 **Appendix A. Supplementary Result Tables**

Table A.5: Baseline performance metrics for the selected bench-
 mark trajectory.

Class	MAE	RMSE	P95 error	Max error	Bias
Direct measurement	0.0311	0.0397	0.0800	1.0000	0.0311
Hybrid direct measurement	0.0024	0.0061	0.0094	1.0000	0.0022
Hybrid ECM	0.0090	0.0127	0.0241	1.0000	-0.0029
Data-driven	0.0100	0.0143	0.0336	0.0686	-0.0062

Table A.6: Cross-scenario global performance metrics used in the result discussion.

Scenario	Class	MAE	Δ MAE	RMSE	Δ RMSE	P95 error
Current noise (high)	DM	0.0310	-0.0001	0.0394	-0.0002	0.0795
Current bias	DM	0.3143	0.2833	0.3683	0.3286	0.6670
Initial SOC error	DM	0.0322	0.0011	0.0413	0.0017	0.0833
Irregular sampling	DM	0.0309	-0.0001	0.0396	-0.0001	0.0799
Burst dropout	DM	0.0332	0.0022	0.0487	0.0090	0.0819
Missing samples	DM	0.0386	0.0076	0.0476	0.0079	0.0926
Voltage spikes	DM	0.0312	0.0001	0.0398	0.0001	0.0801
Temperature noise	DM	0.0311	0.0000	0.0397	0.0000	0.0800
Voltage noise	DM	0.0524	0.0213	0.0648	0.0252	0.1228
Current noise (high)	HDM	0.0044	0.0019	0.0074	0.0013	0.0120
Current bias	HDM	0.3068	0.3043	0.3638	0.3577	0.6668
Initial SOC error	HDM	0.0036	0.0011	0.0136	0.0075	0.0101
Irregular sampling	HDM	0.0027	0.0003	0.0061	0.0001	0.0095
Burst dropout	HDM	0.0051	0.0027	0.0325	0.0265	0.0098
Missing samples	HDM	0.0103	0.0079	0.0129	0.0069	0.0221
Voltage spikes	HDM	0.0062	0.0038	0.0140	0.0079	0.0328
Temperature noise	HDM	0.0067	0.0043	0.0110	0.0049	0.0210
Voltage noise	HDM	0.0751	0.0727	0.0908	0.0847	0.1578
Current noise (high)	HECM	0.0165	0.0075	0.0280	0.0154	0.0701
Current bias	HECM	0.0932	0.0842	0.1303	0.1176	0.2954
Initial SOC error	HECM	0.0091	0.0001	0.0143	0.0016	0.0241
Irregular sampling	HECM	0.0095	0.0004	0.0131	0.0004	0.0248
Burst dropout	HECM	0.0104	0.0014	0.0228	0.0101	0.0246
Missing samples	HECM	0.0119	0.0029	0.0163	0.0036	0.0329
Voltage spikes	HECM	0.0090	0.0000	0.0127	-0.0000	0.0241
Temperature noise	HECM	0.0090	0.0000	0.0127	0.0000	0.0241

Scenario	Class	MAE	Δ MAE	RMSE	Δ RMSE	P95 error
Voltage noise	HECM	0.0090	-0.0000	0.0127	-0.0000	0.0241
Current noise (high)	DD	0.0109	0.0009	0.0151	0.0008	0.0345
Current bias	DD	0.2859	0.2759	0.3426	0.3283	0.6367
Initial SOC error	DD	0.0110	0.0010	0.0175	0.0032	0.0355
Irregular sampling	DD	0.0105	0.0005	0.0144	0.0001	0.0330
Burst dropout	DD	0.0122	0.0022	0.0316	0.0173	0.0348
Missing samples	DD	0.0117	0.0016	0.0149	0.0006	0.0302
Voltage spikes	DD	0.0144	0.0044	0.0213	0.0070	0.0474
Temperature noise	DD	0.0181	0.0081	0.0223	0.0080	0.0451
Voltage noise	DD	0.0698	0.0597	0.0839	0.0696	0.1501

Table A.7: Local behavior and recovery metrics used in the result discussion.

Class	Scenario	Local metric	Value	Threshold
DM	Initial SOC error	Recovery time to strict band [h]	1.1658	0.0800
DM	Initial SOC error	Recovery time to fair band [h]	0.2500	0.1599
HDM	Initial SOC error	Recovery time to strict band [h]	–	0.0094
HDM	Initial SOC error	Recovery time to fair band [h]	–	0.0200
HECM	Initial SOC error	Recovery time to strict band [h]	2.3693	0.0241
HECM	Initial SOC error	Recovery time to fair band [h]	1.1564	0.0482
DD	Initial SOC error	Recovery time to strict band [h]	1.0464	0.0336
DD	Initial SOC error	Recovery time to fair band [h]	1.0014	0.0672
DM	Burst dropout	Recovery time after burst dropout [h]	27.3511	0.0216
HDM	Burst dropout	Recovery time after burst dropout [h]	27.4093	0.0008
HECM	Burst dropout	Recovery time after burst dropout [h]	28.5611	0.0095
DD	Burst dropout	Recovery time after burst dropout [h]	27.4351	0.0070
DM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0800

Class	Scenario	Local metric	Value	Threshold
HDM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0094
HECM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0241
DD	Voltage spikes	Median spike recovery time [s]	0.0000	0.0336
DM	Current noise (high)	Late minus early excess rolling MAE	-0.0013	–
DM	Current noise (high)	Mean excess rolling MAE	-0.0001	–
HDM	Current noise (high)	Late minus early excess rolling MAE	-0.0011	–
HDM	Current noise (high)	Mean excess rolling MAE	0.0019	–
HECM	Current noise (high)	Late minus early excess rolling MAE	0.0038	–
HECM	Current noise (high)	Mean excess rolling MAE	0.0073	–
DD	Current noise (high)	Late minus early excess rolling MAE	-0.0002	–
DD	Current noise (high)	Mean excess rolling MAE	0.0009	–

Table A.8: Scenario overview linking disturbance configuration, primary evidence type, and main paper figure. This table clarifies which scenarios are interpreted predominantly through global metrics and which require additional local evidence.

Scenario	Configuration	Primary evidence	Main figure(s)
ADC quantization	Steps of 0.01 A, 0.005 V, and 0.5 °C	Global Δ MAE; detailed values in Appendix	Figs. 19, B.21
Current bias	Constant current offset in the compact matrix, plus dedicated 0.5%/1.5%/3.0% sweep of $ I _{\max}$	Global MAE degradation across the sweep	Fig. 12

Scenario	Configuration	Primary evidence	Main figure(s)
Current noise	Compact matrix with $\sigma_I = 0.10$ A; local detail sweep up to $\sigma_I = 0.20$ A	Global Δ MAE plus local output-variability / rolling-MAE evidence	Fig. 13
Voltage noise	Additive Gaussian noise with $\sigma_U = 0.01$ V	Global Δ MAE	Fig. 19
Temperature noise	Additive Gaussian noise with $\sigma_T = 1.0$ °C	Global Δ MAE	Fig. 19
Initial SOC error	Additive initial mismatch of -0.10 SOC; DD via online Q_c offset	Local recovery to strict and fair baseline bands; global MAE is secondary	Fig. 14
Missing samples	Every 50th sample frozen on the nominal 1 s grid	Global Δ MAE	Figs. 15, 19
Irregular sampling	Uniform sampling-time jitter of ± 0.1 s around the nominal 1 s grid	Global Δ MAE	Figs. 15, 19
Burst dropout	One central frozen gap of 3600 s	Global Δ MAE plus local transition and long-horizon recovery	Figs. 15, 16, 17
Voltage spikes	One ± 0.20 V single-sample spike every 1000 samples	Global Δ MAE plus aligned local excess-error response	Figs. 18, 19

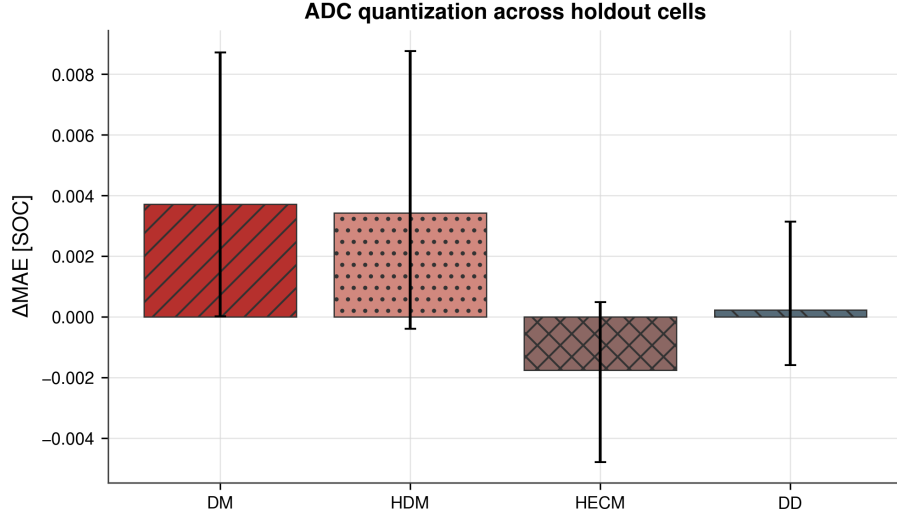


Figure B.21: ADC-quantization results. The two left panels illustrate representative current and voltage quantization in a local window, and the right panel summarizes the resulting global ΔMAE relative to the baseline.

Appendix B. ADC Quantization

ADC quantization is evaluated with step sizes of 0.01 A for current, 0.005 V for voltage, and 0.5 °C for temperature. Across all four estimator classes, the resulting deviations from the baseline remain small, so the detailed numerical comparison is collected here in the appendix while the scenario itself remains part of the overall benchmark.

Table B.9: ADC-quantization results for the four benchmarked estimator classes.

Class	Baseline MAE	ADC MAE	ΔMAE
DM	0.0311	0.0313	+0.0002
HDM	0.0024	0.0029	+0.0005
HECM	0.0090	0.0087	-0.0003
DD	0.0100	0.0098	-0.0002

607 **Funding**

608 Funding details are omitted for double-blind review.

609 **Acknowledgements**

610 Acknowledgements are omitted for double-blind review.

611 **CRedit authorship contribution statement**

612 The author contribution statement is provided in the non-blinded submission
613 materials.

614 **Declaration of competing interest**

615 The authors declare that they have no known competing financial interests or
616 personal relationships that could have appeared to influence the work reported
617 in this paper.

618 **Declaration of generative AI and AI-assisted technologies in the manuscript** 619 **preparation process**

620 During the preparation of this work the author(s) used ChatGPT, DeepL,
621 and Grammarly in order to improve the linguistic quality and style of the
622 manuscript. After using these tools, the author(s) reviewed and edited the con-
623 tent as needed and take(s) full responsibility for the content of the published
624 article.

625 **Data availability**

626 The underlying laboratory dataset is the MGFarm LFP 18650 cyclic-aging
627 dataset described in [28]. The benchmark scripts and model runners are lo-
628 cated in DL_Models/LFP_SOC_SOH_Model/4_simulation_environment,
629 and the paper-facing figures and tables are collected under DL_Models/LFP\

630 _SOC_SOH_Model/4_simulation_environment/results. The data and code
631 underlying this study will be made available on reasonable request.

632 References

- 633 [1] P. Shrivastava, P. A. Naidu, S. Sharma, B. K. Panigrahi, A. Garg, Re-
634 view on technological advancement of lithium-ion battery states estima-
635 tion methods for electric vehicle applications, *Journal of Energy Storage*
636 64 (2023) 107159. doi:10.1016/j.est.2023.107159.
- 637 [2] F. Guo, L. D. Couto, G. Mulder, K. Trad, G. Hu, O. Capron, K. Haghverdi,
638 A systematic review of electrochemical model-based lithium-ion battery
639 state estimation in battery management systems, *Journal of Energy Storage*
640 101 (2024) 113850. doi:10.1016/j.est.2024.113850.
- 641 [3] J. Yao, J. Kowal, Towards a smarter battery management system: A critical
642 review on deep learning-based state of charge estimation of lithium-ion
643 batteries, *Energy and AI* 21 (2025) 100585. doi:10.1016/j.egyai.2025.
644 100585.
- 645 [4] X. Li, J. Jiang, C. Zhang, L. Y. Wang, L. Zheng, Robustness of SOC
646 Estimation Algorithms for EV Lithium-Ion Batteries against Modeling Er-
647 rors and Measurement Noise, *Mathematical Problems in Engineering* 2015
648 (2015) 1–14. doi:10.1155/2015/719490.
- 649 [5] F. Naseri, C. Barbu, A. Jensen, P. Setoodeh, J. R. Baena, N. Hevele,
650 E. Schaltz, Toward Reliable Wireless BMS: Robust Battery State Estima-
651 tion Under Noise and Uncertainty, *IEEE Transactions on Vehicular Tech-*
652 *nology* (2025) 1–12doi:10.1109/TVT.2025.3638613.
- 653 [6] F. Berger, D. Joest, E. Barbers, K. Quade, Z. Wu, D. U. Sauer, P. Dechent,
654 Benchmarking battery management system algorithms - Requirements,
655 scenarios and validation for automotive applications, *eTransportation* 22
656 (2024) 100355. doi:10.1016/j.etrans.2024.100355.

- 657 [7] Z. A. Khan, P. Shrivastava, S. M. Amrr, S. Mekhilef, A. A. Algethami,
658 M. Seyedmahmoudian, A. Stojcevski, A Comparative Study on Different
659 Online State of Charge Estimation Algorithms for Lithium-Ion Batteries,
660 Sustainability 14 (12) (2022) 7412. doi:10.3390/su14127412.
- 661 [8] Y. Gao, G. L. Plett, G. Fan, X. Zhang, Enhanced state-of-charge estimation
662 of LiFePO₄ batteries using an augmented physics-based model, Journal
663 of Power Sources 544 (2022) 231889. doi:10.1016/j.jpowsour.2022.
664 231889.
- 665 [9] Y. Gao, K. Liu, C. Zhu, X. Zhang, D. Zhang, Co-Estimation of State-of-
666 Charge and State-of- Health for Lithium-Ion Batteries Using an Enhanced
667 Electrochemical Model, IEEE Transactions on Industrial Electronics 69 (3)
668 (2022) 2684–2696. doi:10.1109/TIE.2021.3066946.
- 669 [10] J. Shen, W. Ma, J. Xiong, X. Shu, Y. Zhang, Z. Chen, Y. Liu, Alterna-
670 tive combined co-estimation of state of charge and capacity for lithium-
671 ion batteries in wide temperature scope, Energy 244 (2022) 123236. doi:
672 10.1016/j.energy.2022.123236.
- 673 [11] C. Wang, N. Cui, Z. Cui, H. Yuan, C. Zhang, Fusion estimation of lithium-
674 ion battery state of charge and state of health considering the effect of
675 temperature, Journal of Energy Storage 53 (2022) 105075. doi:10.1016/
676 j.est.2022.105075.
- 677 [12] P. Mondal, D. Bhavsar, K. Mittal, M. Mittal, Estimating State-of-Charge
678 in Lithium-Ion Batteries Through Deep Learning Techniques: A Com-
679 parative Evaluation, IEEE Access 12 (2024) 78773–78786. doi:10.1109/
680 ACCESS.2024.3408220.
- 681 [13] D. P. Pau, A. Aniballi, Tiny Machine Learning Battery State-of-Charge
682 Estimation Hardware Accelerated, Applied Sciences 14 (14) (2024) 6240.
683 doi:10.3390/app14146240.

- [14] S. Giazitzis, M. Sakwa, S. Leva, E. Ogliari, S. Badha, F. Rosetti, A Case Study of a Tiny Machine Learning Application for Battery State-of-Charge Estimation, *Electronics* 13 (10) (2024) 1964. doi:10.3390/electronics13101964.
- [15] Y. Mazzi, H. Ben Sassi, A. Gaga, F. Errahimi, State of charge estimation of an electric vehicle's battery using tiny neural network embedded on small microcontroller units, *International Journal of Energy Research* 46 (6) (2022) 8102–8119. doi:10.1002/er.7713.
- [16] M. A. Kamali, A. C. Caliwag, W. Lim, Novel SOH Estimation of Lithium-Ion Batteries for Real-Time Embedded Applications, *IEEE Embedded Systems Letters* 13 (4) (2021) 206–209. doi:10.1109/LES.2021.3078443.
- [17] B. Gou, Y. Xu, X. Feng, An Ensemble Learning-Based Data-Driven Method for Online State-of-Health Estimation of Lithium-Ion Batteries, *IEEE Transactions on Transportation Electrification* 7 (2) (2021) 422–436. doi:10.1109/TTE.2020.3029295.
- [18] W. Li, N. Sengupta, P. Dechent, D. Howey, A. Annaswamy, D. U. Sauer, Online capacity estimation of lithium-ion batteries with deep long short-term memory networks, *Journal of Power Sources* 482 (2021) 228863. doi:10.1016/j.jpowsour.2020.228863.
- [19] F. Rzepka, P. Hematty, M. Schmitz, J. Kowal, Neural Network Architecture for Determining the Aging of Stationary Storage Systems in Smart Grids, *Energies* 16 (17) (2023) 6103. doi:10.3390/en16176103.
- [20] BQ79616 16-Series Battery Monitor, Balancer, and Integrated Hardware Protector (2023).
- [21] Design Considerations of Current Sensing With BQ769x2 Family for Battery Management System (2022).
- [22] TMCS1108 3% Functional Isolation Hall-Effect Current Sensor With ± 100 -V Working Voltage (2024).

- 712 [23] TLE4972 high precision coreless current sensor for automotive applications
713 (2023).
- 714 [24] O. Aiello, Electromagnetic Susceptibility of Battery Management Systems’
715 ICs for Electric Vehicles: Experimental Study, *Electronics* 9 (3) (2020) 510.
716 doi:10.3390/electronics9030510.
- 717 [25] M. Liu, D. Geißler, S. Bian, B. Zhou, P. Lukowicz, Assessing the Impact of
718 Sampling Irregularity in Time Series Data: Human Activity Recognition
719 As A Case Study (Jan. 2025). arXiv:2501.15330, doi:10.48550/arXiv.
720 2501.15330.
- 721 [26] 10s battery pack monitoring, balancing, and comprehensive protection, 50-
722 A discharge reference design (2024).
- 723 [27] A. Aqachtoul, K. Karam, A. Elamrani, Y. Alidrissi, M. Najib, A. Rochd,
724 M. Bakhouya, MGFARM: An IoT and Edge-Driven Framework for Smart
725 and Sustainable Agricultural Practices, in: Y. Mejdoub, A. Elamri, M. Kar-
726 douchi (Eds.), *Connected Objects, Artificial Intelligence, Telecommunica-*
727 *tions and Electronics Engineering*, Vol. 1584, Springer Nature Switzerland,
728 Cham, 2026, pp. 419–425. doi:10.1007/978-3-032-01536-5_64.
- 729 [28] F. Rzepka, Design of Experiments for Cyclic Aging of LFP 18650 Cells: Im-
730 pact of Charge/Discharge C-Rates, Depth of Discharge, and Temperature
731 on Battery Degradation (Nov. 2025). doi:10.14279/DEPOSITONCE-24800.
- 732 [29] K. Siebertz, D. Van Bebber, T. Hochkirchen, *Statistische Versuchspla-*
733 *nung*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2017. doi:10.1007/
734 978-3-662-55743-3.